Exam review

Friday, November 22, 2024 6:58 PM

Least squares

curve fitting

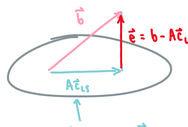
$A\vec{x} \approx \vec{b}$

$A\vec{x} = \vec{b} \Rightarrow \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$

$A\vec{x} = x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}$

find vector $\vec{x}_{LS}$ closest to $\vec{b}$, which does not lie in vector space of matrix A

error matrix: $\vec{e} = \vec{b} - A\vec{x}_{LS}$



projection of $\vec{b}$ onto column space of A

find this $\vec{y}_{LS} = A\vec{x}_{LS}$ that gets us closest to $\vec{b}$ (minimize $\|\vec{e}\|$)

find $\vec{x}_{LS}$ s.t. $\vec{e} = \vec{b} - A\vec{x}_{LS} \perp$ to col(A)

$\vec{e} = \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_m \end{bmatrix}$ $\vec{e} \perp$ col(A) $A = \begin{bmatrix} | & | & & | \\ \vec{a}_1 & \vec{a}_2 & \cdots & \vec{a}_n \\ | & | & & | \end{bmatrix}$ $A^T = \begin{bmatrix} — \vec{a}_1^T — \\ — \vec{a}_2^T — \\ \vdots \\ — \vec{a}_n^T — \end{bmatrix}$

$\vec{e} \cdot \vec{a}_1 = 0$

$\vec{e} \cdot \vec{a}_2 = 0$

$\vdots$

$\vec{e} \cdot \vec{a}_n = 0$

$A^T \cdot \vec{e} = \begin{bmatrix} — \vec{a}_1^T — \\ — \vec{a}_2^T — \\ \vdots \\ — \vec{a}_n^T — \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_m \end{bmatrix} = \begin{bmatrix} \vec{a}_1 \cdot \vec{e} \\ \vec{a}_2 \cdot \vec{e} \\ \vdots \\ \vec{a}_n \cdot \vec{e} \end{bmatrix} = \vec{0}$

$A^T(\vec{b} - A\vec{x}_{LS}) = \vec{0}$

$A^T\vec{b} - A^TA\vec{x}_{LS} = 0$

$A^TA\,\vec{x}_{LS} = A^T\vec{b}$

A^* — n×m m×1

$b^* = A^T\vec{b}$ Let $A^* = A^TA$

$A^*\vec{x}_{LS} = \vec{b}^*$ $\vec{b}^* = A^T\vec{b}$

solve with Gaussian elimination

$A^TA\,\vec{x}_{LS} = A^T\vec{y}$

e.g. fit (x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4) to quadratic $y = c_1 + c_2x + c_3x^2$

$\begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \\ 1 & x_4 & x_4^2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix}$

$A^T = \begin{bmatrix} 1 & 1 & 1 & 1 \\ x_1 & x_2 & x_3 & x_4 \\ x_1^2 & x_2^2 & x_3^2 & x_4^2 \end{bmatrix}$

e.g. $(-1, 6)$, $(2, 0)$, $(3, 2)$, fit as linear $a + bx$

$\begin{bmatrix} 1 & -1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \\ 2 \end{bmatrix}$

$A^*\vec{x} = A^TA = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$

$= \begin{bmatrix} 3 & 4 \\ 4 & 14 \end{bmatrix}$

$\begin{bmatrix} 3 & 4 \\ 4 & 14 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 6 \\ 0 \\ 2 \end{bmatrix}$ $y = c_1 + c_2x$

$= \begin{bmatrix} 8 \\ 0 \end{bmatrix}$

$\left[\begin{array}{cc|c} 3 & 4 & 8 \\ 4 & 14 & 0 \end{array}\right]$

$R_1/3$:

$\left[\begin{array}{cc|c} 1 & 4/3 & 8/3 \\ 4 & 14 & 0 \end{array}\right]$

$R_2 - 4R_1$:

$\left[\begin{array}{cc|c} 1 & 4/3 & 8/3 \\ 0 & 22/3 & -32/3 \end{array}\right]$

$R_2 / \frac{22}{3}$

$\left[\begin{array}{cc|c} 1 & 4/3 & 8/3 \\ 0 & 1 & -16/11 \end{array}\right]$

$R_1 - \frac{4}{3}R_2$:

$\left[\begin{array}{cc|c} 1 & 0 & 56/11 \\ 0 & 1 & -16/11 \end{array}\right]$

$c_1 = \frac{56}{11} = 4.81$

$c_2 = -\frac{16}{11} = -1.23$

$y = 4.31 - 1.23x$

RREF

$\left[\begin{array}{cccc} 1 & 3 & 0 & 2 \\ 0 & 0 & 1 & 4 \\ 1 & 3 & 1 & 6 \end{array}\right] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 6 \\ 6 \end{bmatrix}$

$\left[\begin{array}{cccc|c} 1 & 3 & 0 & 2 & 1 \\ 0 & 0 & 1 & 4 & 6 \\ 1 & 3 & 1 & 6 & 6 \end{array}\right]$

$R_3 - R_1$:

$\left[\begin{array}{cccc|c} 1 & 3 & 0 & 2 & 1 \\ 0 & 0 & 1 & 4 & 6 \\ 0 & 0 & 1 & 4 & 5 \end{array}\right]$

$R_3 - R_2$:

$\left[\begin{array}{cccc|c} 1 & 3 & 0 & 2 & 1 \\ 0 & 0 & 1 & 4 & 6 \\ 0 & 0 & 0 & 0 & -1 \end{array}\right]$

no sol

RREF: $\left[\begin{array}{cccc} 1 & 3 & 0 & 2 \\ 0 & 0 & 1 & 4 \end{array}\right]$

① first nonzero entry in each row is 1

② all other entries in col w/ leading 1 is 0

③ 1's move right as you go down

④ all zeros collected @ bottom

Types of solutions to elimination

↳ unique solution: all leading, no free

↳ infinite: at least one free variable

↳ no sol: [0 0 0 | non zero]

r should be # of leading 1's

compare rank(A) (# of leading (1's) to m (rows) or n (cols)

↳ if r<n: some cols are free vars → infinite sols

↳ if r=n: all cols are independent → one sol

↳ if r<m: may be inconsistent (if [0 0 0 | ≠0])

↳ r=m: always consistent

Square systems

$\begin{bmatrix} m & p \\ n & q \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$

r=1: $\left[\begin{array}{cc|c} 1 & p/m & b_1/m \\ 0 & 0 & d_2 \end{array}\right]$ $d_2 = 0$: inf. sols; $d_2 \neq 0$: no sols

r=2: $\left[\begin{array}{cc|c} 1 & 0 & b_1/m \\ 0 & 1 & d_2 \end{array}\right]$ unique sol

Tall and thin systems m>n

$\begin{bmatrix} m & p \\ n & q \\ p & h \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$

r=1: $\left[\begin{array}{cc|c} 1 & p/m & b_1/m \\ 0 & 0 & d_2 \\ 0 & 0 & d_3 \end{array}\right]$ (free var)

$d_2 \neq 0$ or $d_3 \neq 0$: no sol

$d_2 = d_3 = 0$: inf. sols (1 free variable)

r=2: $\left[\begin{array}{cc|c} 1 & 0 & d_1 \\ 0 & 1 & d_2 \\ 0 & 0 & d_3 \end{array}\right]$

$d_3 \neq 0$: no sol

$d_3 = 0$: unique sol

Short and wide systems m<n

$\begin{bmatrix} m & p & f \\ n & q & g \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$

r=1: $\left[\begin{array}{ccc|c} 1 & p/m & f/m & b_1/m \\ 0 & 0 & 0 & d_1 \end{array}\right]$ $d_1 \neq 0$: no sol; $d_1 = 0$: inf. sols

r=2: $\left[\begin{array}{ccc|c} 1 & 0 & f/m & b_1/m \\ 0 & 1 & g/q & b_2/q \end{array}\right]$ inf. sols

(leading vars; free var)

Only matrix not full rank has 0 or inf. sols

Gaussian Elimination

e.g. $\begin{cases} x + 2y + z = 2 \\ 3x + 8y + z = 12 \\ 4y + z = 2 \end{cases}$

$A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix}$

forward rep: eliminate all numbers below pivot

$R_2 - 3R_1$:

$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{bmatrix}$

$R_3 - 2R_2$:

$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 5 \end{bmatrix}$ warning: pivot cannot be 0, otherwise swap rows

$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 2 & -2 & 6 \\ 0 & 0 & 5 & -10 \end{array}\right]$

backward rep: all other entries above pivot are also 0

(similar operations but reverse order)

$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 2 & -2 & 6 \\ 0 & 0 & 1 & -2 \end{array}\right]$

$R_2 + 2R_3$: $\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 2 & 0 & 2 \\ 0 & 0 & 1 & -2 \end{array}\right]$

$R_1 - R_3 - R_2$: $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 2 & 0 & 2 \\ 0 & 0 & 1 & -2 \end{array}\right]$

$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -2 \end{array}\right]$ ← RREF

$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$

Matrix properties

① $A + B = B + A$

② $c(A+B) = cA + cB$

③ $A + (B+C) = (A+B) + C$

④ $C(A+B) = CA + CB$

⑤ $(A+B)C = AC + BC$

⑥ $(AB)C = A(BC)$

Inverse matrices

For a square matrix, A (n×n),

A is invertible if there exists another matrix (A^{-1}) such that

$AA^{-1} = I$

$A^{-1}A = I$

If $A\vec{x} = \vec{b}$, then $A^{-1}\vec{b} = \vec{x}$ ⇒ implies there is only 1 soln.

Finding the inverse:

$TT^{-1} = I$

If $T = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $T^{-1} = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$, then

$\begin{bmatrix} e & f \\ g & h \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

expand: $T^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

determinant: cannot equal 0

Determinant: $\det(T) = ad - bc$

Inverse with Gaussian Elimination

$[A|I]$

$[\underbrace{E_kE_{k-1}\cdots E_2E_1A}_{I} | \underbrace{E_kE_{k-1}\cdots E_2E_1I}_{A^{-1}}]$

if A is invertible, $E_kE_{k-1}\cdots E_2E_1A = I \Rightarrow E_kE_{k-1}\cdots E_2E_1 = A^{-1}$

$A = (E_kE_{k-1}\cdots E_2E_1)^{-1} = E_1^{-1}E_2^{-1}\cdots E_k^{-1}$

properties:

① can have max 1 inverse

② If A is invertible, there is 1 solution to $A\vec{x} = \vec{b}$

③ If A is invertible, only sol to $A\vec{x} = \vec{0}$ is $\vec{x} = 0$

④ Invertible only if RREF is identity matrix

⑤ Product of AB invertible only if A and B both are

elimination matrices with inverse matrix:

E^{-1} is opposite row operation

e.g. $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$R_2 - 5R_1$ ↓ $R_2 + 5R_1$ ↓

$E = \begin{bmatrix} 1 & 0 & 0 \\ -5 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $E^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 5 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$E^{-1}E = \begin{bmatrix} 1 & 0 & 0 \\ -5 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 5 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$

e.g. $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$R_2 = A - R_1 \cdot c$:

$\begin{bmatrix} a & b \\ 0 & ad-bc \end{bmatrix}$

if $ad - bc = 0$ ⇒ zero row ⇒ A not invertible

determinant

$\det(A) = 0$: not invertible

$\det(A) \neq 0$: invertible

Initial Value Problems

Euler's method

$\begin{bmatrix} y_{n+1} \\ y'_{n+1} \end{bmatrix} = \begin{bmatrix} y_n \\ y'_n \end{bmatrix} + \Delta t\, A \begin{bmatrix} y_n \\ y'_n \end{bmatrix}$

state matrix

$\begin{bmatrix} y' \\ y'' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 4 & -3 \end{bmatrix} \begin{bmatrix} y \\ y' \end{bmatrix}$

$y'' = -3y' + 4y$

OR

$t_{n+1} = t_n + \Delta t$

$y_{n+1} = y_n + \Delta t\, \underbrace{f(t_n, y_n)}_{y'}$ $(y'(t) = f(t, y(t)))$

Improved Euler's method

average of two slopes

$t_{n+1} = t_n + \Delta t$

$y_{n+1} = y_n + \Delta t \left(\frac{f(t_n, y_n) + f(t_{n+1}, y_{n+1})}{2} \right)$ (estimate from EM)

OR

$y_{n+1} = y_n + \frac{\Delta t}{2}(Ay_n + Ay_{n+1})$

$\begin{bmatrix} y_{n+1} \\ y'_{n+1} \end{bmatrix} = \begin{bmatrix} y_n \\ y'_n \end{bmatrix} + \frac{\Delta t}{2}\left(A\begin{bmatrix} y_n \\ y'_n \end{bmatrix} + A\left(\begin{bmatrix} y_n \\ y'_n \end{bmatrix} + \Delta t A \begin{bmatrix} y_n \\ y'_n \end{bmatrix}\right)\right)$

Boundary Value Problems

Finite difference method

① Divide into n evenly spaced intervals from x=a to x=b

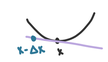
step size: $\Delta x = \frac{b-a}{n}$

② Approximate difference of each interval as a differential equation

- Forward difference $\Delta_F f(x) = \frac{f(x+\Delta x) - f(x)}{\Delta x}$



- Backward difference $\Delta_B f(x) = \frac{f(x) - f(x-\Delta x)}{\Delta x}$



- Central difference $\Delta_C f(x) = \frac{f(x+\Delta x) - f(x-\Delta x)}{\Delta x}$



can combine methods to approximate y''

$f''(x) \approx \Delta_B[\Delta_F f(x))]$

$\approx \frac{\Delta_F f(x) - \Delta_F f(x-\Delta x)}{\Delta x}$

$= \frac{f(x+\Delta x) - f(x) - f(x) + f(x-\Delta x)}{(\Delta x)^2}$

$= \frac{f(x+\Delta x) - 2f(x) + f(x-\Delta x)}{(\Delta x)^2}$

③ Solve system of n-1 equations using linear algebra